SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**
Question Count: 5

Duration :60 Minutes

Code of Conduct
I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student: ____________________

Assessment Tasks Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low

latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would implement.

5. **Agriculture Smart Farming Problem**
   A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

1. Retail Data Optimization Problem

Introduction

A retail company has physical stores and an online platform. It generates large amounts of **sales, inventory, customer, and online browsing data**. A big data architecture is required for real-time inventory tracking, demand forecasting, and personalized recommendations.

1. Data Collection

Data can be collected from:

- Point-of-sale (POS) systems
- Online shopping websites
- Mobile applications
- Inventory sensors
- Customer interactions

Apache Kafka can be used to collect real-time events.

2. Data Storage

A data lake using HDFS or cloud storage can store large volumes of raw data.

Structured sales and inventory data can be stored in relational databases or data warehouses, while customer clickstream data can be stored in NoSQL/data-lake storage.

3. Real-Time Inventory Tracking

Kafka can send inventory events to Apache Spark Streaming or Apache Flink.

For example:

Sale → Kafka → Stream Processing → Inventory Database → Updated Stock

This allows the company to know current stock levels.

4. Demand Forecasting

Historical sales data can be processed using **Apache Spark**.

Machine learning models such as:

- Linear Regression
- Random Forest
- Time-series forecasting

can predict future product demand.

5. Personalized Recommendations

Customer browsing and purchase history can be analyzed using recommendation algorithms such as:

- Collaborative filtering
- Content-based filtering
- Machine learning models

For example, customers who purchase a mobile phone may receive recommendations for phone cases and chargers.

6. Scalability

The system can use:

- Distributed storage
- Kafka partitions
- Database sharding
- Horizontal scaling
- Data replication

This allows the platform to handle increasing amounts of retail data.

2. Social Media Fraud Detection Problem

Introduction

A social media platform processes billions of posts, comments, likes, and shares. Fake accounts and misinformation must be detected quickly. Therefore, a combination of **big data, stream processing, and machine learning** is required.

1. Data Collection

User activities such as posts, comments, likes, shares, login details, and account information are collected continuously.

Apache Kafka can be used as the distributed event-streaming system.

2. Distributed Storage

Large amounts of historical data can be stored in:

- HDFS
- Cloud data lakes
- NoSQL databases

The data can be partitioned by user, date, region, or event type.

3. Stream Processing

Apache Flink or Spark Streaming can process user activities in real time.

Example:

User Activity → Kafka → Flink/Spark → Fraud Detection Model → Alert

This allows suspicious accounts to be detected quickly.

4. Machine Learning

Machine learning can identify unusual behavior.

Useful features include:

- Number of posts
- Posting frequency
- Number of followers
- Login locations
- Sharing patterns
- Account age
- Similarity between posts

Algorithms such as **Random Forest, Logistic Regression, and clustering** can be used.

5. Fake Account Detection

A fake account may have:

- Very high activity
- Few genuine connections
- Repeated content
- Unusual login patterns
- Automated behavior

ML models can assign a **fraud probability score**.

6. Misinformation Detection

Natural Language Processing (NLP) can analyze text to identify suspicious or misleading content. Content can be compared with trusted sources and previous patterns.

7. Scalability and Fault Tolerance

The system can use:

- Kafka partitions
- Distributed processing
- Replication
- Load balancing
- Multiple processing nodes

This allows billions of events to be processed.

3. Government Data Platform Problem

Introduction

A government data platform combines census, economic, geographic, and other departmental data. Since many departments need to share information, the platform requires strong **data governance, interoperability, security, and privacy**.

1. Data Collection

Data can come from:

- Census systems
- Government departments
- Economic databases
- Geographic information systems
- Public services

Data may be structured, semi-structured, or unstructured.

2. Centralized Data Lake

A government data lake can store large amounts of information using **HDFS or cloud object storage**.

Sensitive structured information can also be maintained in secure relational databases.

3. Data Integration

ETL tools can combine information from different departments.

The process is:

Extract → Transform → Clean → Integrate → Store

Common data formats and APIs can help departments exchange information.

4. Interoperability

A common data model and standard APIs should be used so that different government systems can communicate.

Metadata and data catalogs should describe:

- Data source
- Data owner
- Data format
- Data quality
- Update frequency

5. Data Governance

A governance framework should define:

- Data ownership
- Data quality rules
- Access policies
- Retention policies

- Data standards

A data governance team can monitor compliance.

6. Privacy and Security

Government data should be protected using:

- Encryption
- Authentication
- Role-based access control
- Data masking
- Secure APIs
- Audit logs

Only authorized departments should access sensitive information.

7. Data Quality

Data should be checked for:

- Missing values
- Duplicate records
- Incorrect formats
- Inconsistent information

Data validation improves reliability.

4. Banking Fraud Detection Problem

Introduction

Banks process millions of transactions every minute. Fraud must be detected in real time while maintaining low latency and high accuracy. A combination of **stream processing, big data storage, and machine learning** can solve this problem.

1. Transaction Data Collection

Transactions contain information such as:

- Account number
- Transaction amount
- Location
- Time
- Merchant
- Payment method

Apache Kafka can collect transaction events continuously.

2. Real-Time Processing

Apache Flink or Spark Streaming can analyze transactions immediately.

Example:

Transaction → Kafka → Flink → Fraud Model → Decision

If a transaction is suspicious, it can be blocked or sent for verification.

3. Historical Data

Past transactions can be stored in:

- Data lakes
- HDFS
- Data warehouses
- Distributed databases

Historical data is used for training fraud detection models.

4. Machine Learning Algorithms
Possible algorithms include:
- Logistic Regression
- Random Forest
- Gradient Boosting
- Neural Networks
- Isolation Forest for anomaly detection

The model produces a fraud probability score.
5. Fraud Features
Important features include:
- Unusual transaction amount
- Unusual location
- Multiple transactions in a short period
- New device
- Unusual spending pattern
- Multiple failed transactions

6. Rule-Based Detection
Rules can also be used along with ML.
For example:
If transaction amount is unusually high AND location is unusual → Flag transaction
Combining rules and ML improves detection.
7. Low Latency and Scalability
To achieve fast processing:
- Kafka partitions distribute transactions.
- Stream processors run in parallel.
- Data is cached when required.
- Multiple servers provide scalability.
- Replication provides fault tolerance.

5. Agriculture Smart Farming Problem
Introduction
Smart farming collects data from **soil sensors, weather stations, drones, satellites, and crop monitoring systems**. Big data analytics can help farmers improve crop yield and reduce water, fertilizer, and other resource usage.
1. Data Collection
Sensors can collect:
- Soil moisture
- Soil temperature
- Humidity
- Nutrient levels
- Weather conditions

Drones and satellites can provide crop images and field information.
2. Data Ingestion
IoT gateways can collect sensor data and send it to the central platform.
Sensors/Drones → IoT Gateway → Kafka → Data Platform

Kafka can handle continuous sensor data.

3. Data Storage

A cloud data lake or HDFS can store large volumes of:

- Sensor data
- Weather records
- Drone images
- Crop information

Time-series databases can efficiently store sensor readings.

4. Real-Time Analytics

Spark Streaming or Flink can analyze sensor data in real time.

For example, if soil moisture becomes too low, the system can immediately recommend or trigger irrigation.

5. Machine Learning

ML models can predict:

- Crop yield
- Disease occurrence
- Irrigation requirements
- Pest attacks
- Suitable planting periods

Historical weather and crop data can be used to train these models.

6. Image Analysis

Drone images can be analyzed using computer vision to identify:

- Diseased crops
- Dry areas
- Pest damage
- Poor plant growth

7. Resource Optimization

Analytics can help determine the correct amount of:

- Water
- Fertilizer
- Pesticides
- Seeds

This reduces costs and prevents wastage.